

Project Title: UK Train Rides

Description:

This project focuses on analyzing UK Train Rides data using Microsoft Power BI. The goal is to extract meaningful insights from sales, routes, station usage, and ticket patterns to support data-driven decision-making for railway operations. Through visual dashboards and interactive reports, this project identifies trends, peak travel times, and performance metrics of railway services. The data was processed, cleaned, and visualized to provide actionable insights for business intelligence.

Tools Used:

- Microsoft Power BI
- CSV Data
- Data Cleaning & DAX Measures

Key Points:

- Railway Ticketing Performance Overview
- Revenue & Pricing Analysis
- Customer Behavior & Purchase Patterns
- Booking Channel & Time-of-Purchase Analysis
- Route-Level Performance Analysis
- Journey Performance Analysis
- Refund Requests & Delay Impact Analysis

Railway Ticketing Performance Overview

1. Revenue and Ticket Sales

- **Total Revenue:** 741,900
- **Tickets Sold:** 31,700
- **Journeys Completed:** 19,100
- **Refund Requests:** 1,120
- **Average Ticket Price:** 23.44

Although many tickets were sold, a smaller number of journeys were completed. This could mean some passengers didn't use their tickets or had them refunded. It's worth checking why some tickets are not being used.

2. Ticket Sales by Month

- **January:** 8,434
- **February:** 7,337
- **March:** 8,100
- **April:** 7,748

January had the most ticket sales, while February had the lowest. This could be due to fewer travel days or weather. Knowing these trends helps with planning promotions and resources during busy and slow months.

3. Where People Buy Tickets

- **Online:** 18,500 tickets (58.5%)
- **At the Station:** 13,100 tickets (41.5%)

Most people prefer to buy tickets online. Making online ticketing even easier and faster could increase sales. However, physical stations are still important for some travelers, so support should continue there too.

4. Ticket Classes

- **Standard Class:** 28,600 tickets
- **First Class:** 3,100 tickets

Most passengers choose standard class, likely because it's cheaper. To earn more, promotions could be offered to encourage more people to try first class (like discounts or free upgrades).

5. Payment Methods

- **Credit Card:** 19,136
- **Contactless:** 10,834
- **Debit Card:** 1,683

People mostly pay with credit cards or contactless options. Debit card use is much lower. This shows that fast, digital payments are preferred, so adding more contactless and mobile options may be useful.

6. Most Common Departure Stations

1. Manchester Piccadilly
2. London Euston
3. Liverpool Lime Street
4. London Paddington
5. London Kings Cross

These stations are the most used for departures, likely because they are in large or busy cities. More staff, clearer signs, or better facilities in these locations could improve the passenger experience.

7. Most Common Arrival Destinations

1. Birmingham New Street
2. Liverpool Lime Street
3. York
4. Manchester Piccadilly
5. Reading

These cities are popular travel destinations, either for work or tourism. Special offers or services targeting these routes could boost sales even more.

8. Journey Status

- **On Time:** 18,019
- **Delayed:** 1,062
- **Cancelled:** 790

Most trains run on time, which is great. But over 1,800 journeys were delayed or cancelled. This affects customer satisfaction, so it's important to reduce these issues.

9. Why Trains Were Delayed (and Refunds)

Cause of Delay	Refund Requested	No Refund Requested
Bad Weather	133	407
Not Enough Staff	248	268
Signal Problems	204	334
Technical Issues	305	137
Traffic on Tracks	117	82

Most delays were due to staffing and technical issues, and many people asked for refunds because of them. Fixing these problems can improve service and reduce refunds.

Revenue & Pricing Analysis

1. Revenue Summary

- **Total Money Earned (Before Refunds): \$741.9K**
- **Money Kept (After Refunds): \$703.2K**
- **Refunds Cost: \$38.7K**

- **Tickets Sold: 31,700**
- **Average Ticket Price: \$23.44**
- **Ticket Price Range: Lowest \$1 – Highest \$267**

2. Monthly Revenue

- **January: \$194K (highest)**
- **February: \$146K (lowest)**
- **March: \$184K**
- **April: \$178K**

Sales were best in January and lowest in February. This may be due to fewer people traveling or bad weather. Knowing this helps plan offers slower months.

3. Most Profitable Train Routes

The top 5 routes that made the most money:

- 1. London Kings Cross → York**
- 2. Liverpool Lime Street → London Euston**
- 3. London Paddington → Reading**
- 4. London Euston → Manchester Piccadilly**
- 5. London Street → Birmingham New Street**

These routes bring in the most income. Improving services on these routes could increase revenue even more.

4. Most Profitable Railcard

- **Adult Railcard Revenue: \$70,730**

Most of the money from railcards comes from adults. Offering deals or loyalty points to adult passengers could help bring in even more sales.

5. Refund Causes

Reasons for refunds and how much they cost:

- **Technical Issues: \$15K**
- **Staff Shortages: \$12K**
- **Track Traffic: \$5K**
- **Signal Problems: \$4K**
- **Weather Delays: \$3K**

Most refunds happened because of technical or staffing problems. Fixing these issues could reduce costs and improve customer satisfaction.

6. Revenue by Ticket Class

- **Standard Class Revenue: \$113.6K**
- **First Class Revenue: \$33.9K**
- **Average Price Per Ticket:**
 - **First Class: \$48.86**
 - **Standard Class: \$20.72**

Most passengers buy standard tickets, but first-class tickets earn more per passenger. Offering discounts or upgrades could help sell more first-class tickets.

Customer Behavior & Purchase Patterns

1. Railcard Usage

- **Top Railcard:** *Adult Railcard*
- **Other Popular Railcards:** Disabled, Senior
- **Railcard Holders:**
 - Adult: 4,846 users
 - Disabled: 3,089 users
 - Senior: 2,800 users
- **Passengers without a Railcard:** 20,918 users
- **Overall Railcard Usage Rate:** 33.9%

Most passengers do not use a railcard, but a third of them do. Promoting railcards, especially to frequent travelers, could increase sales and loyalty.

2. Railcard Revenue Over Time

Adult Railcard Revenue

- Jan: \$20K
- Feb: \$15K
- Mar: \$18K
- Apr: \$17K

Disabled Railcard Revenue

- Jan: \$13.6K
- Feb: \$9.85K
- Mar: \$13.8K
- Apr: \$13.7K

Senior Railcard Revenue

- Jan: \$7.3K
- Feb: \$5.3K
- Mar: \$6.6K
- Apr: \$6.6K

Adult railcards generate the most consistent and highest revenue. Disabled and senior cards bring less revenue but still play a key role in accessibility and customer satisfaction.

3. How People Buy Tickets

- **Online:** 58.5%
- **At Station:** 41.5%

More than half of tickets are bought online. Improving the online booking experience can further increase sales, but stations should still provide strong service for those who prefer in-person purchases.

4. Ticket Classes

- **First Class Tickets:** 9.7% of sales
- **Standard Class (implied):** 90.3%

Most people choose standard class, but nearly 1 in 10 use first class. Running promotions, like weekend upgrades, could encourage more use of first class.

5. Most Common Ticket Types

- **Top Ticket Type:** Advance
- **Other Types:** Off-Peak and Anytime
- **Transactions by Type:**

- Advance: 17,561
- Off-Peak: 8,752
- Anytime: 5,340

Advance tickets are the most popular. These are often cheaper, so highlighting them in marketing could help increase early bookings.

7. Railcard and Ticket Type Combinations

Breakdown of how railcard holders use ticket types:

- **No Railcard Users:**
 - Advance: 36.7%
 - Off-Peak: 18.7%
 - Anytime: 10.6%
- **Adult Railcard Users:**
 - Advance: 8.6%
 - Off-Peak: 4.18%
 - Anytime: 2.56%
- **Senior Railcard Users:**
 - Advance: 4.94%
 - Off-Peak: 1.98%
 - Anytime: 1.92%

Most users without a railcard buy advance tickets. Users with railcards also prefer advance fares, but off-peak and anytime tickets are also used. These patterns show that railcards encourage flexible travel choices.

Booking Channel & Time-of-Purchase Analysis

1. Booking Overview

- **Total Bookings:** 31,700
- **Average Booking Time:** 1.66 days before the journey
- **Most Bookings:** Made close to the travel date

Passengers usually book only 1–2 days in advance. This suggests a high rate of last-minute or urgent travel. Offering better pricing or flexibility for early bookings may encourage people to plan ahead.

2. Booking Channels

- **Online Bookings:** 58.5%
- **Station Bookings:** 41.5%

More than half of the tickets are bought online. This highlights the need for a smooth and user-friendly digital experience, while still supporting passengers who book at stations.

3. Delay Refund Rates by Booking Channel

- **Online Refund Rate for Delays:** 0.2%
- **Station Refund Rate for Delays:** 3.8%
- **Average Delay Causing Refund:** 2.1 minutes

Refunds due to delays are much more frequent for station bookings than online bookings. This may suggest differences in customer expectations or how delay information is communicated. Improving communication at stations could help reduce refund requests.

4. Booking Channel by Railcard Type

Online Purchases:

- None: 64.31%
- Senior: 11.99%
- Adult: 11.95%
- Disabled: 11.75%

Station Purchases:

- None: 68.59%
- Adult: 20.05%
- Disabled: 6.95%
- Senior: 4.41%

Most people without railcards book both online and at stations. Adult railcard holders prefer station booking more than others. Targeted promotions online could shift more railcard users toward digital channels.

5. Ticket Types by Booking Window

Booking windows are grouped by how early tickets are bought:

- **Last-Minute (<1 day):**
 - Anytime: 5,000

- **Immediate (1–5 days):**
 - Advance: 15,000
- **Short-Term (6–10 days):**
 - ~1,000 bookings
- **Planned (11–17 days):**
 - ~1,000 bookings
- **Early Booking (>17 days):**
 - 568 bookings

Most people buy **advance tickets** 1 to 5 days before travel. Very few book more than 10 days in advance. This shows strong short-term demand, and there is room to promote early-bird discounts to balance sales and improve planning.

Route-Level Performance Analysis

1. Overall Route Activity

- **Total Journeys:** 19,900
- **Number of Routes:** 65
- **Departure Stations:** 12
- **Arrival Destinations:** 32
- **Average Ticket Price:** \$23.44
- **Average Delay Time:** 37.73 minutes

This shows a wide variety of routes, but also indicates that delays are common, with an average delay of over half an hour. That may impact customer satisfaction and should be investigated.

2. Routes with the Highest Delay Rates (100% Delayed)

- **Edinburgh Waverley → London Kings Cross**
- **London Euston → York**
- **York → Wakefield**

These routes had delays on every journey. Immediate investigation is needed to identify whether this is due to infrastructure, scheduling, or external factors.

3. Routes with the Highest Cancellation Rates

- **Liverpool Lime Street → Birmingham New Street:** 14.3% cancelled
- **Reading → Didcot Parkway:** 11.4%

- **London Paddington → Oxford:** 11%

These routes are experiencing frequent cancellations, likely frustrating passengers and increasing refund costs. Identifying causes like train availability, demand, or track issues is recommended.

4. Routes with the Highest Ticket Prices

- **Manchester Piccadilly → London Paddington:** \$114
- **Liverpool Lime Street → London Stratford:** \$105
- **Liverpool Lime Street → London Euston:** \$103

These are premium routes with high prices, possibly due to longer distances or express service. Ensuring top service quality here is important to justify the price.

5. Routes with the Longest Delays

- **Manchester Piccadilly → Leeds:** 138 minutes
- **York → Doncaster:** 69 minutes
- **Manchester Piccadilly → Liverpool Lime Street:** 68 minutes

These routes have extreme average delays. The Manchester–Leeds route in particular needs urgent review to reduce waiting times and improve reliability.

6. Routes with the Most Refund Requests

- **Liverpool Lime Street → London Euston:** 171 refunds
- **Liverpool Lime Street → Manchester Piccadilly:** 116 refunds
- **Birmingham New Street → Manchester Piccadilly:** 102 refunds

High refund numbers on these routes suggest service issues or high expectations not being met. These should be a priority for quality improvement efforts.

Journey Performance Analysis

1. Overall Journey Performance

- **On-Time Rate:** 90.7%
- **Delay Rate:** 5.3%
- **Cancellation Rate:** 3.98%

Most journeys run on time, which is positive. However, about **9% of journeys** are either delayed or cancelled, which still impacts many passengers.

2. Monthly Trends – Delays and Cancellations

Month	Cancelled	Delayed
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January	207	266
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February	181	256
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March	215	294
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April	187	246
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March saw the **highest number of delays and cancellations**, suggesting possible seasonal or operational factors that should be explored further.

3. Average Delay Time by Cause

Reason	Avg Delay (min)
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Signal Failure	61.09
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Staffing Issues	42.14
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Weather Conditions	35.04
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Technical Problems	20.93
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Track Traffic	20.44
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Signal failures cause the longest delays, averaging over an hour. Reducing signal-related issues could have a major impact on punctuality.

4. Busiest Departure Stations

Station	Journeys Departed
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Manchester Piccadilly	3,520
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London Euston	3,060
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Liverpool Lime Street	3,000
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These are the most frequently used starting points. Consistent service quality at these stations is essential.

5. Delay Times by Departure Station

Station	Avg Delay (min)
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Manchester Piccadilly	54
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London Euston	46
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Station	Avg Delay (min)
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York	36
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Manchester Piccadilly has the highest average delay. Targeted improvements here (e.g., staffing, scheduling) could greatly reduce overall delay rates.

Refund Requests & Delay Impact Analysis

1. Monthly Refund Requests

- **January:** 149 requests
- **February:** 133 requests
- **March:** 168 requests (*highest*)
- **April:** 150 requests

Refund requests are fairly steady, with a slight increase in March, possibly due to the high number of delays that month.

2. Refunds by Journey Status

- **Cancelled Journeys:** 572 refund requests
- **Delayed Journeys:** 546 refund requests

Refunds are almost equally requested for both delays and cancellations, highlighting the need to reduce both types of disruptions.

3. Delay Causes by Journey Status

Cause	Cancelled	Delayed
Weather	186	280
Staffing Issues	192	245
Signal Failure	207	217
Technical Issue	105	268
Traffic	100	52

Weather and technical issues are major causes of delays, while **signal failures** are a common reason for cancellations.

4. Refund Likelihood by Delay Cause

Cause	% Not Refunded	% Refunded
Weather	90.31%	9.69%
Signal Failure	77.84%	22.16%
Staffing Issues	67.74%	32.26%
Technical Issue	45.12%	54.88%
Traffic	61.46%	38.54%

Technical issues have the highest chance of triggering a refund. This suggests they significantly affect service quality and customer expectations.

5. Refund Requests by Railcard Ownership

Railcard Type	No Refund	Refund
None	97.53%	2.47%
Adult	91.29%	8.71%
Disabled	98.06%	1.94%
Senior	95.71%	4.29%

Passengers without railcards still make up most refund requests. This might reflect their higher numbers or greater dissatisfaction due to lack of benefits.